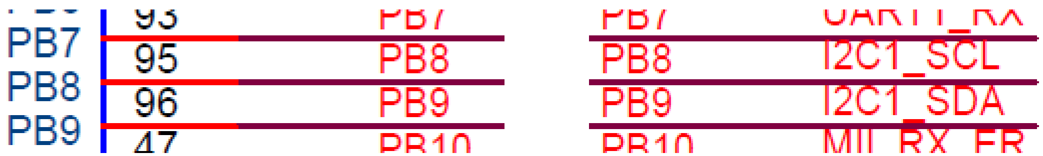
I2C - BMA280 테스트

CRZ Technology

Mango-M32F2 I2C-BMA280 3-Axis sensor 테스트

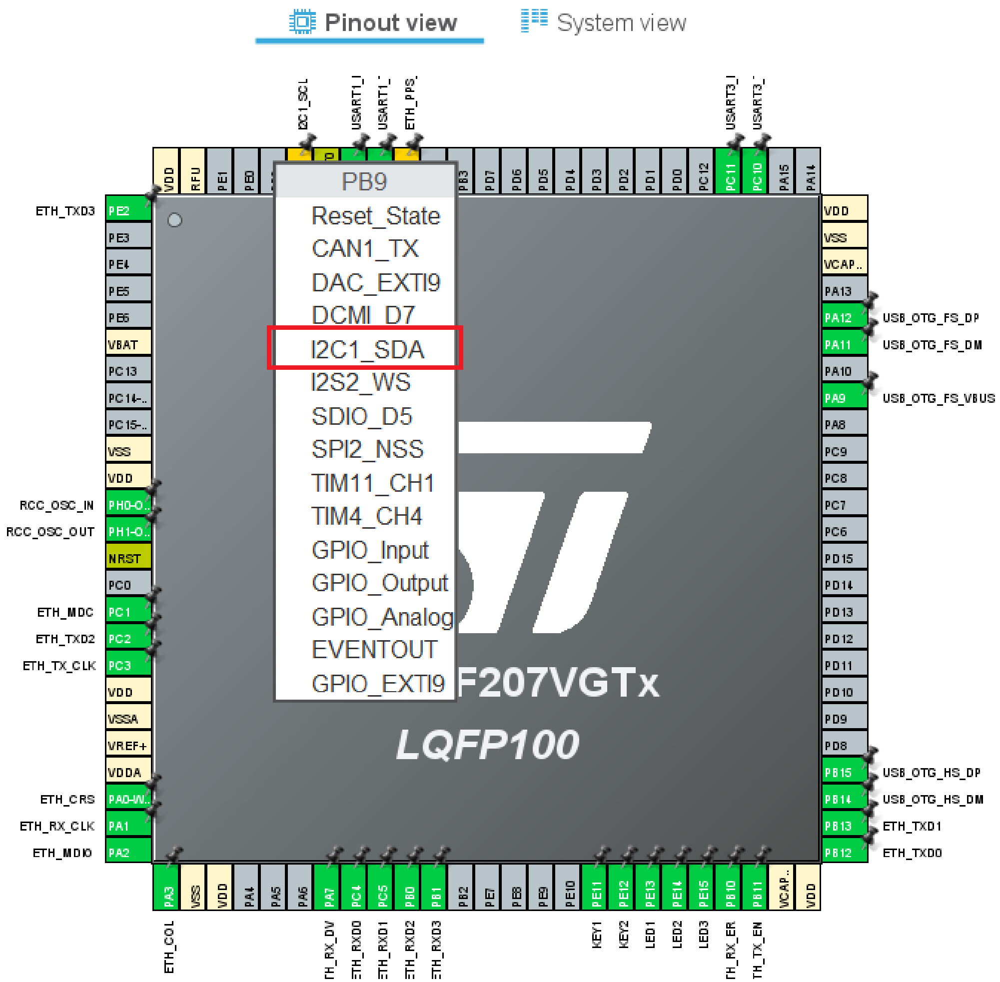
I2C pinmap

- **I2C – PB8, PB9**



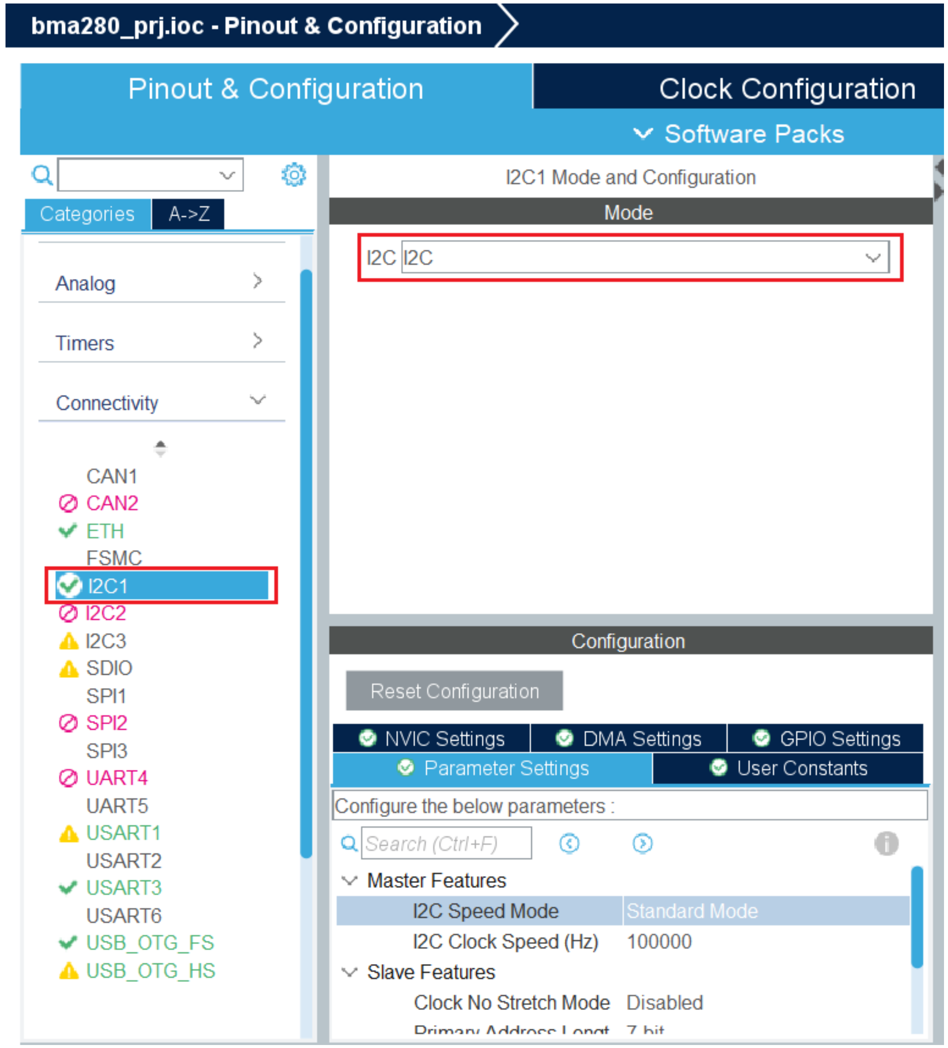
Pinout view

- **PB8-I2C1_SCL, PB9-I2C1_SDA 선택.**



I2C 설정

- “Connectivity“ 클릭, “I2C1” 클릭.
- “Mode“ - “I2C” 클릭.



BMA280 Test Project

- **Code 생성 후 main.c 수정**

```c
/* Private define ------------------------------------------------------------*/
/* USER CODE BEGIN PD */

#define BMA280_I2C_ADDR (0x18 << 1)

#define BMA280_RA_CHIP_ID           0x00
#define BMA280_RA_X_AXIS_LSB        0x02
#define BMA280_RA_BANDWIDTH       0x10
#define BMA280_RA_RANGE_BWIDTH      0x0f

/* range and bandwidth */
#define BMA280_RANGE_2G                0x3
#define BMA280_RANGE_4G                0x5
#define BMA280_RANGE_8G                0x8

#define BMA280_BW_7HZ                 0
#define BMA280_BW_7_81HZ                 0x8
#define BMA280_BW_15_13HZ                0x9
#define BMA280_BW_125HZ                12
#define BMA280_BW_250HZ                13
#define BMA280_BW_500HZ                14
#define BMA280_BW_MAX              0xF

/* USER CODE END PD */
```

BMA280 Test Project

```
/* USER CODE BEGIN 0 */
static unsigned int count = 0;

static uint8_t Read_1_Byte(uint8_t address)
{
    uint8_t Buffer_Tx;
    uint8_t Buffer_Rx;

    Buffer_Tx = address;

    HAL_I2C_Master_Transmit(&hi2c1, BMA280_I2C_ADDR, &Buffer_Tx, 1, 1000);
    HAL_I2C_Master_Receive(&hi2c1, BMA280_I2C_ADDR, &Buffer_Rx, 1, 1000);

    return Buffer_Rx;
}

static uint8_t Get_ChipID(void)
{
    return Read_1_Byte(BMA280_RA_CHIP_ID);
}

/*
 * Description : none.
 * Argument(s) : none.
 * Return(s)   : none.
 * Note(s)     : none.
 */
static void Read_3_Axis_BMA280_Data(int16_t * xData_p, int16_t * yData_p, int16_t * zData_p)
{
    uint8_t Buffer_Tx;
    uint8_t Buffer_Rx[6];

    Buffer_Tx = BMA280_RA_X_AXIS_LSB;

    HAL_I2C_Master_Transmit(&hi2c1, BMA280_I2C_ADDR, &Buffer_Tx, 1, 1000);
    HAL_I2C_Master_Receive(&hi2c1, BMA280_I2C_ADDR, Buffer_Rx, 6, 1000);

    * xData_p = ((((int16_t)(int8_t)Buffer_Rx[1]) << 8) | Buffer_Rx[0]) >> 6;
    * yData_p = ((((int16_t)(int8_t)Buffer_Rx[3]) << 8) | Buffer_Rx[2]) >> 6;
    * zData_p = ((((int16_t)(int8_t)Buffer_Rx[5]) << 8) | Buffer_Rx[4]) >> 6;
}
/* USER CODE END 0 */
```

BMA280 Test Project

```c
  MX_I2C1_Init();
  /* USER CODE BEGIN 2 */
  printf("I2C-BMA280 test\r\n");
  httpd_init();

  printf("Get_ChipID: %X\n", Get_ChipID());

  Buffer_Tx[0] = BMA280_RA_RANGE_BWIDTH;
  Buffer_Tx[1] = BMA280_RANGE_8G << 3;
  HAL_I2C_Master_Transmit(&hi2c1, BMA280_I2C_ADDR, Buffer_Tx, 2, 1000);
  Buffer_Tx[0] = BMA280_RA_BANDWIDTH;
  Buffer_Tx[1] = BMA280_BW_500HZ;
  HAL_I2C_Master_Transmit(&hi2c1, BMA280_I2C_ADDR, Buffer_Tx, 2, 1000);
  printf("Set Param BMA280_RANGE_8G, BMA280_BW_500HZ\n");
  /* USER CODE END 2 */

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    MX_LWIP_Process();

    if(count++ % 10000 == 0)
    {
        Read_3_Axis_BMA280_Data(&xData, &yData, &zData);
        printf("x: %d, y: %d, z: %d\n", xData, yData, zData);
    }
    /* USER CODE END WHILE */
    MX_USB_HOST_Process();

    /* USER CODE BEGIN 3 */
  }
  /* USER CODE END 3 */
}
```

Test 결과 확인

- **CR-zF-BMA280 센서 보드를 8Pin Cable을 이용 CON7에 연결함.**
- **Chip ID – 0xFB 정상적으로 읽힘.**
- **센서를 움직일 때 3-axis 데이터가 변동함.**

COM4 - Tera Term VT

File Edit Setup Control Window Help

```
I2C-BMA280 test
Get_ChipID: FB
Set Param BMA280_RANGE_8G, BMA280_BW_500HZ
x: 162, y: 180, z: -53
x: 163, y: 217, z: -282
x: 121, y: 247, z: -213
x: 26, y: 454, z: -344
x: 82, y: 421, z: 21
x: 511, y: 220, z: 511
x: -51, y: 156, z: 32
x: -186, y: 511, z: -319
x: -63, y: -118, z: 189
x: -16, y: 511, z: 27
x: 511, y: 472, z: 511
x: 349, y: 415, z: 148
x: -158, y: 48, z: 4
x: -172, y: 17, z: 25
x: 171, y: 237, z: 172
x: 272, y: 445, z: 511
x: 33, y: 258, z: 91
x: 323, y: 222, z: 173
x: 306, y: 264, z: 51
x: 238, y: 150, z: 44
```

Thank You